Supplementary Table 3 Infection risk in School setting

Source	First author	Main issue	Study type	Population (N)	N primary/ index cases	N primary cases staff	N primary cases student	N Contact staff	N contacts student	N contacts	N tested	Setting	Comparator	Date	Outcome	N tested positive	Overall risk of infection	Infection risk in teacher	Infection risk in student	Comments	Infection dynamics[1]	Incident cases and deaths per day[2]	Control measures
Infection risk in School																							
Public Health Agency Sweden	Public Health Agency Sweden [31]	Risk of SARS-CoV-2 infection in teachers in Sweden	Report (observational study)	Teachers working in different school in Sweden	364760	479 cases	NA	NA	NA	NA	NA	School of Sweden	Teachers of different school forms	March-May 2020	Relative risk	**NA**		**RR and 95%CI** High School teachers **0.7(0.5-1)** Primary school teachers **1.1(0.9-1.3)** Leisure educators **0.8(0.5-1.3)** Preschool teachers **0.7(0.6-0.9)** Other educators with theoretical specialists	NA	The relative risk is higher for primary school teachers and student assistants compared to other groups working in schools in Sweden	3,800.47 cases per million 435.08 deaths per million	**Incidence** Start= 0 Peak= 80.01 End= 26.24 **Death** Start= 0.1 Peak= 18.32 End= 0	Partial school closure

[1] Total number of COVID-19 cases and deaths from COVID-19 per million during the period of data collection/study duration in the country of study
[2] The number of incidence case per million and number of death per million per day. The data presented are at the start of the study period, peak during the study period and at the end of the study period

																	1(0.7-1.5) Nanny **1(0.8-1.2)** Student assistants and other **1.1(0.8-1.4)**					
medRvix	Vlachos,J [33]	Risk of COVID-19 infection in the parents, high school teachers, lower secondary school teacher and their partners	Retrospective data analysis	Upper secondary and lower secondary teaches, their partners and student's parents from Sweden	NA	NA	NA	NA	NA	NA	NA	School and household of Sweden	Teachers of different school form, their partners And parents of students	June 15, 2020	Odds ratio and 95%CI	**NA**	**lower secondary school** **parents** **1.15 (1.03-1.27)** **Partners** **1.30(1.00-1.68)**	**2.01(1.52 – 2.67)**	NA	Secondary school is the reference group	2,833.37 cases per million 92.09 deaths per million	**Incidence** Start= 64.16 Peak= 168.13 End= 79.19 **Deaths** Start= 0.79 Peak= 10.69 End= 5.25	Partial school closure
Emerging Infectious Diseases	Brown, NE[26]	Teacher to student transmission	Observational (Serological survey)	121 (1+120)	1	1	NA	NA	120	120	21	School in USA.	Students	Blood sample collection date: 13 March, 2020	Infection rate	**2**	**1.66%**	**0**	**1.66%**	blood collection was done 14 days after exposure	587.87 cases per million 16.13 deaths per	**Incidence** Start= 0.02 Peak= 79.50 End= 79.50 **Death**	NA

																				million	Start= 0 Peak= 3.31 End= 3.31	
medRvix	Lopez, L [37]	Sero-prevalence of anti-SARS-CoV-2 IgG antibodies in school staff in Midwestern USA	Population based sero prevalence study	1261 eligible School staff	NA	NA	NA	NA	NA	NA	753 participated in the survey (Annual check-up)	18 year or older staff of Lake Central School Corporation located in suburban Indiana	General population	July 2020	Seroprevalence of antibody	22 participants tested positive for SARS-CoV-2 antibodies	2.9% (95%CI 1.8-4.4%) **When Corrected for performance parameter** 1.7%	1.7%		The seroprevalence of LCSC staff is 1.7% which is below the seroprevalence reported in metropolitan areas throughout the USA	5,658.69 cases per million 78.18 deaths per million	**Incidence** Start= 156.52 Peak= 229.03 End= 207.54 **Death** Start= 2.14 Peak = 4.33 End= 3.72	School closure, While entering the building for data collection facemask, social distancing and hand sanitizer were mandatory
medRvix	Armann, JP [25]	Seroprevalence study in student and teacher to assess their role SARS-CoV-2 transmission	Seroprevalence study	2045 (1538 students and 507 teachers)	NA	NA	NA	NA	NA	NA	2045	Secondary school in Saxony, Germany	students	25 May, 2020 and 30 June, 2020	Prevalence (Seroprevalence)	12 (11 students and 1 teacher)	0.6%	0.2%	0.7%	Seroprevalence study in 13 different secondary school in Saxony, Germany	374.07 cases per million 26.90 deaths per million	**Incidence** Start= 12.75 Peak= 15.13 End= 4.49 **Death** Start= 1.35	School closure, face mask and social distancing

																					Peak= 3.37 End= 0.17	
medRvix	Fontanet, A [28]	to assess the infection attack rate using serological assays	Retrospective cohort study (Seroprevalence study)	661 (including 240 pupil, 211 parents, 127 sibling, 53 teacher; 27 non-teaching staff and 3 other)	NA	NA	NA	NA	NA	NA	NA	661	Secondary school and household	Student, parents	30 March -4 April, 2020	Infection attack rate	171 (pupil= 92; teacher= 23; school staff= 16; parents=24; sibling = 13, and other= 3)	25.9% (22.6-29.4)	12.18%	28.75%	Seroprevalence study in secondary school and household in Oise in Northern France	41.81 cases per million 69.49 deaths per million	**Incidence** Start= 66.70 Peak= 116.88 End=13.10 **Death** Start= 6.40 Peak= 19.18 End= 16.13	NA
medRvix	Lynda, F[41]	Risk of COVID-19 and risk of hospitalization with COVID-19 among teachers compared with healthcare workers and general population of working age	Case control study	871568	NA	NA	NA	NA	NA	NA	NA	School, hospital and household of Scotland	Healthcare worker, Household member of healthcare worker and teacher	March 2020-January 2021	Risk rate/ odds ratio	NA		**Before re-opening**: Any cases 0.48 (0.39-0.60); Hospitalization 0.51 (0.31-0.82) **School re-opened:** Any cases 1.42	**Before reopening,** **Any case** HH_Teacher 0.78 (0.66-0.91) HCW 8.75 (8.29-9.23); HH_HCW 2.32 (2.11-2.56) **Hospitalization** HH_Teacher 0.63	comparators are healthcare workers and household member of teachers and healthcare worker The results presented are adjusted ratios obtained	56389.96 cases per million 1566.83 deaths per million	**Incidence** Start= 0.03 Peak= 1004.51 End= 312.14 **Death** Start= 0.01 Peak= 26.90 End= 17.75	School close and again reopen

																		(1.35-1.49); Hospitalization 0.97 (0.72-1.29)	(0.44-0.91); HCW 3.15 (2.71-3.66); HH_HCW 1.87 (1.52-2.30) **Re-opened** **Any cases** HH_Teacher 1.01 (0.95-1.06); HCW 2.41 (2.33-2.49); HH_HCW 1.25 (1.20-1.31) **Hospitalization** HH_Teacher 0.97 (0.74-1.27); HCW 1.82 (1.55-2.14), HH_HCW 1.04 (0.83-1.23)	from logistic regression			